Hydronium ions inhibit CO_2 reduction on coinage metals

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The carbon-efficient electrochemical reduction of CO_2 (CO_2RR) requires acidic electrolytes to mitigate irreversible electrolyte carbonation. However, CO_2RR only takes place in acidic electrolytes upon addition of inert spectator cations and the mechanistic basis for this requirement remains unclear. Herein, we present scanning voltammetry experiments conducted in bulk acidic electrolytes in conjunction with dynamic electrochemical mass spectrometry (DEMS) analysis of CO_2 consumption and product formation. By quantifying both faradaic CO_2 consumption via electroreduction and non-faradaic CO_2 consumption via electrolyte carbonation in real-time (**Figure 1**), we deconvolute the direct role of the added electrolyte cations from their indirect role in fostering a transient local pH swing. We find that across coinage metals and electrolyte compositions, CO_2RR commences in parallel with non-faradaic CO_2 uptake via carbonation, indicating that interfacial alkalinization is pre-requisite for, rather than merely a consequence of CO_2RR. Doping Ag with Pt to promote hydronium consumption via H_2 evolution serves to lower the overpotential for CO_2RR onset in scanning voltammetry experiments by ~1 V (**Figure 2**). These findings evince that the primary role of the supporting electrolyte cations is to establish a non-equilibrium interfacial ionic environment depleted in hydronium ions and implies a putative mechanistic model in which hydronium ions serve to inhibit CO_2 activation at coinage metal surfaces. This

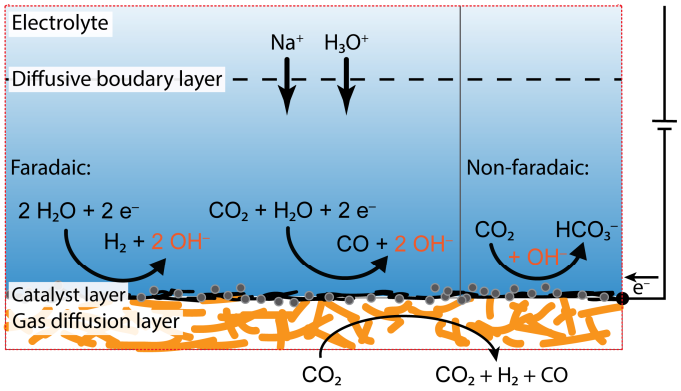

Figure 1. An expanded view of the gas diffusion electrode used in this study is shown with the different Faradaic and non-Faradaic reactions occurring during electrolysis. In addition, the ion transport components across the diffusive boundary layer and the gradient in pH are outlined.

work highlights the value of pre-steady state real-time analysis of both faradaic and non-faradaic interfacial reactions and establishes a self-consistent mechanistic picture for CO_2RR in bulk acidic electrolyte.

Discussion

From all the data we obtained, it becomes obvious that the ionic environment at the electrode|electrolyte interface is a key determinant of both CO_2RR and HER rates. The onset potential of CO formation on coinage metal electrodes during LSV cycles in bulk acid seems to be determined by the change in local pH rather than a set overpotential. At a high level, this leads us to invoke a pH-dependent process with a net electrosorption valency close to 0. We therefore invoke a mechanistic model for CO_2RR catalysis in which a pre-rate-limiting step for the reaction is influenced by the presence of a surface species exhibiting both (1) pH-dependent; and (2) potential-independent coverage. We can envision two different scenarios, one in which increases in pH deplete pre-associated hydronium at the interface leading to the liberation of CDR sites. Alternatively, anions that form at the electrode|electrolyte interface during the interfacial pH change might be beneficial for the formation of CO_2RR products.

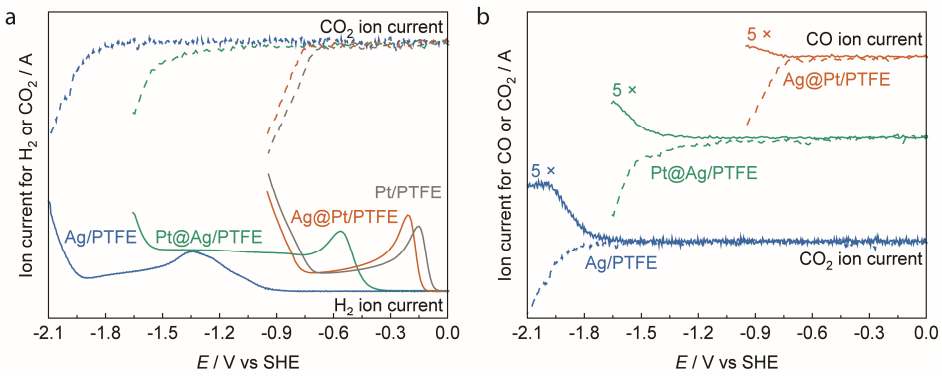

Figure 2. Interfacial pH swings induced by Pt-catalyzed HER lead to drastically lower onset potentials of CO_2 reduction on Ag/PTFE. a) H_2 and CO_2 ion currents during CV cycles in a 100 mM $HClO_4$ / 900 mM $NaClO_4$ electrolyte. b) CO_2 and CO ion currents during CV cycles in a 100 mM $HClO_4$ / 900 mM $NaClO_4$ electrolyte. Pt@Ag/PTFE samples were prepared by depositing 1 nm Pt on top of 300 nm Ag sputtered on PTFE membranes. Similarly, Ag@Pt/PTFE samples contain 1 nm Ag sputtered on top of 300 nm Pt/PTFE.

Significance

This work highlights that CO_2 reduction on metal electrodes in bulk acid electrolytes only occurs after an increase of interfacial pH. In addition to important mechanistic insight, we provide a lower bound for the thermodynamic concentration overpotential penalty that needs to be paid for running CO_2 electrolysis in acidic electrolytes.